

MATH 74: ALGEBRAIC TOPOLOGY

PRISHITA DHARAMPAL

1. TOPOLOGICAL SPACES & HOMEOMORPHISMS

Definition 1.1. A **topology** on a set X is a collection J of subsets of X having the following properties.

- (1) $\emptyset, X \in J$
- (2) The union of the elements of any sub-collection of J is in J .
- (3) The intersection of finitely many elements in J is in J .

A set for which a topology has been specified is said to be a topological space. We think of it as a pair (X, J) .

Definition 1.2. Elements of J are called **open sets** of the topological space (X, J) .

Definition 1.3. A subspace Y of a topological space (X, J) is a subset $Y \subset X$ which is given the **subspace topology** where the open sets of $Y = \{U \cap Y : U \in J\}$

Definition 1.4. A space X is **path connected** if every pair of points of X can be joined by a path in X .

Definition 1.5. Let (X_1, J_1) and (X_2, J_2) be topological spaces. A map $f : (X_1, J_1) \rightarrow (X_2, J_2)$ is said to be **continuous** if $f^{-1}(U) \in J_1$ for all $U \in J_2$.

Definition 1.6. Let (X, J) and (X', J') be topological spaces. Then they are **homeomorphic** if $\exists$ a continuous bijection $f : (X, J) \rightarrow (X', J')$. So that f^{-1} is also continuous.

Exercise 1.1. Show that any two open intervals of $\mathbb{R}$ are homeomorphic (where we put subspace topology on the open intervals).

Solution. Let (a, b) and (c, d) be open intervals. **TODO:**

Remark 1.7. Number of path connected topologies is preserved by continuous maps. For instance, $[0, 1]$ is not homeomorphic to $(0, 1)$, and $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^2$.

Exercise 1.2.

- (a) Is $\mathbb{R}^2$ homeomorphic to $\mathbb{R}^3$?
- (b) Is $S^2 (= x^2 + y^2 + z^2 = 1)$ homeomorphic to $T^2 = S^1 \times S^1$?
- (c) Is S^2 homeomorphic to $S^1 \times [a, b]$?

Solution.

- (a) **TODO:**

Definition 1.8. If f and f' are continuous maps from spaces X to Y we say that f is homotopic to f' if $\exists$ a continuous map $F : X \times I = [0, 1] \rightarrow Y$ such that $F(x, 0) = f(x)$ and $F(x, 1) = f'(x)$. F is called the **homotopy** between f and f' , and we write $f \simeq f'$.

Definition 1.9. If X and Y are topological spaces, we can define a **product topology** on $X \times Y$ by saying that a basis consists of the subsets $U \times V$ as U ranges over open sets in X and V ranges over open sets in Y .

Definition 1.10. Define a **path** f as $f : [0, 1] \rightarrow X$ with $f(0) = x_0$ and $f(1) = x$.

Definition 1.11. Two paths $f, f' : [0, 1] \rightarrow X$ are said to be path homotopic if there exist a continuous map $F : I \times I \rightarrow X$ such that $\forall s \in I$,

$$F(s, 0) = f(s), \quad F(s, 1) = f'(s)$$

and $\forall t \in I$,

$$F(0, t) = x_0, \quad F(1, t) = x$$

Then we write $f \simeq_p f'$.

Example 1.1. Let $f : X \rightarrow \mathbb{R}^2$ and $g : X \rightarrow \mathbb{R}^2$ then $f \simeq g$.

Proof. Want to find F such that $F : X \times I \rightarrow \mathbb{R}^2$. $F(x, 0) = f(x)$, $F(x, 1) = g(x)$ for all $x \in X$. The following map (straight line homotopy) works.

$$F(x, t) = (1 - t)f(x) + tg(x)$$

□

Lemma 1.12. The relations $\simeq$ and $\simeq_p$ are equivalence relations.

Proof.

Homotopy is an equivalence relation:

- (1) **Reflexive:** $f \simeq f$ by $F(s, t) = f(s)$.
- (2) **Symmetric:** Let $f \simeq g$ by the homotopy $F(s, t)$. Then $g \simeq f$ is homotopic by the homotopy $G(s, t) = F(s, 1 - t)$.
- (3) **Transitive:** Let $f \simeq g$ by $F(s_1, t)$ and $g \simeq h$ by $G(s_2, t)$. We can show that $f \simeq h$ by $H(s, t) = \begin{cases} s = s_1, 0 \leq t \leq 1/2 \\ s = s_2, 1/2 \leq t \leq 1 \end{cases}$ **TODO: check and also prove for $\simeq_p$**

Thus path-homotopies are equivalence relations and we write $[f]$ to denote the path-homotopy equivalence class of f . □

Lemma 1.13 (Pasting Lemma). Let $X = A \cup B$ be a topological space with $A, B \subset X$ are closed. Let $f : A \rightarrow Y, g : B \rightarrow Y$ be continuous and $f(x) = g(x), \forall x \in A \cap B$, then

$$h(x) = \begin{cases} f(x), & x \in A \\ g(x), & x \in B \end{cases}$$

is continuous.

Definition 1.14. Let $f : I \rightarrow X$ be a path from x_0 to x_1 and $g : I \rightarrow X$ be a path from x_1 to x_2 , then we define a new path

$$f \star g : I \rightarrow X$$

from x_0 to x_2 by

$$f \star g(t) = \begin{cases} f(2t), & t \in [0, 1/2] \\ g(2t - 1), & t \in [1/2, 1] \end{cases}$$

where $\star$ is **composition of paths**.

Some Properties of ' $\star$ '.

- (1) ' $\star$ ' is well-defined operation in the path homotopy equivalence class,

$$[f] \star [g] = [f \star g]$$

- (2) Identity:

$$[f] \star [e_{x_1}] = [f], \quad [e_{x_0}] \star [f] = [f]$$

where e_{x_t} is the constant path based at $x \in X$ and $t \in [0, 1]$

- (3) Inverse:

$$[f] \star [\bar{f}] = [e_{x_0}], \quad [\bar{f}] \star [f] = [e_{x_1}]$$

where $\bar{f}(t) = f(1 - t)$.

- (4) Associative:

$$[f] \star ([g] \star [h]) = ([f] \star [g]) \star [h]$$

Proof.

- (1) Let $f \simeq_p f'$ by F and $g \simeq g'$ by G , then we can show that $f \star g \simeq f' \star g'$ by defining $H : I \times I \rightarrow X$ such that

$$H(s, t) = \begin{cases} F(2s, t), & s \in [0, 1/2] \\ G(2s - 1, t), & s \in [1/2, 1] \end{cases}$$

- (2) $f \star e_{x_1} \simeq f$ by $H : I \times I \rightarrow X$ such that

$$H(s, t) = \begin{cases} f\left(\frac{2s}{2-t}\right), & s \in [0, \frac{2-t}{2}] \\ x_1, & s \in [\frac{2-t}{2}, 1] \end{cases}$$

- (3) $f \star \bar{f} \simeq_p e_{x_0}$.

Define a path $\alpha(t)$ that starts at x_0 and ends at $f(t)$. Then the path-homotopy is given by $\alpha(t) \star \bar{\alpha}(t)$.

$$H(s, t) = \begin{cases} f(2st), & s \in [0, 1/2] \\ f(2t(1 - s)), & s \in [1/2, 1] \end{cases}$$

TODO: straight line homotopy

- (4) **TODO:**

□

Remark 1.15. Note that $\star$ operation on the set of path-homotopy equivalence classes of paths in the space X forms a groupoid structure.

2. THE FUNDAMENTAL GROUP

Definition 2.1. Let X be a space and $x_0 \in X$ be a point in X we say that a path $f : I \rightarrow X$ is a **loop** based at x_0 if $f(0) = f(1) = x_0$

Definition 2.2. The set of all path-homotopy equivalence classes of loops based at x_0 forms a group under $\star$ called the **fundamental group** of X based at x_0 , denoted as $\pi_1(X, x_0)$

- Identity: $[e_{x_0}]$
- Inverses: $[f] \rightarrow [\bar{f}]$

Example 2.1. $\pi_1(\mathbb{R}^n, (0,0)) \equiv 0 \equiv \langle e_{origin} \rangle$

Proof. Take any two loops $f(t), g(t)$ based at the origin. Then consider the straight-line homotopy

$$H(s, t) = f(s)(1 - t) + tg(s)$$

TODO: (Check that it is a path homotopy) □

Definition 2.3. Let α be a path from x_0 to x_1 , where $x_0, x_1 \in X$. We define a map $\hat{\alpha} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ such that

$$\hat{\alpha}([f]) := [\bar{\alpha}] \star [f] \star [\alpha] = [\bar{\alpha} \star f \star \alpha]$$

Proposition 2.4. $\hat{\alpha}$ is well-defined and an isomorphism.

Proof. Let $f \simeq_p f'$. We need to check that

$$\begin{aligned} [\bar{\alpha}] \star [f] \star [\alpha] &= [\bar{\alpha}] \star [f'] \star [\alpha] \\ \iff [\bar{\alpha} \star f \star \alpha] &= [\bar{\alpha} \star f' \star \alpha] \\ \iff \bar{\alpha} \star f \star \alpha &\simeq_p \bar{\alpha} \star f' \star \alpha \end{aligned}$$

Since $\star$ is well-defined, $[f] \star [\alpha] = [f'] \star [\alpha]$, $\hat{\alpha}$ is well-defined

Let $[f], [g] \in \pi_1(X, x_0)$.

$$\begin{aligned} \hat{\alpha}([f]) \star \hat{\alpha}([g]) &= [\bar{\alpha}] \star [f] \star [\alpha] \star [\bar{\alpha}] \star [g] \star [\alpha] \\ &= [\bar{\alpha}] \star [f] \star [e_{x_0}] \star [g] \star [\alpha] \\ &= [\bar{\alpha}] \star [f] \star [g] \star [\alpha] \\ &= \hat{\alpha}([f] \star [g]) \end{aligned}$$

Hence $\hat{\alpha}$ is a homomorphism. To check that $\hat{\alpha}$ is an isomorphism, we see that $\hat{\alpha}(\hat{\alpha}([f])) = [f]$ for all $[f] \in \pi_1(X, x_0)$. □

Corollary 2.4.1. The fundamental group of X based at x_0 only depends on the path-connected component of $x_0 \in X$ upto isomorphism.

Definition 2.5. Let $h : X \rightarrow Y$ be a continuous map such that $h(x_0) = y_0$, $x_0 \in X, y_0 \in Y$. Then define $h_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ by $h_*([f]) = [h \circ f]$. This is a **push-forward** map.

Proposition 2.6. h_* is a well-defined homomorphism.

Proof. Let $f \simeq_p f'$. We need to check that $h \circ f \simeq_p h \circ f'$. Let $H : I \times I \rightarrow X$ be the path homotopy between f and f' . Consider $h \circ H : I \times I \rightarrow Y$.

$$\begin{aligned} h \circ H(s, 0) &= h \circ f \\ h \circ H(s, 1) &= h \circ f' \\ h \circ H(0, t) &= h \circ f(0) \\ h \circ H(1, t) &= h \circ f(1) = h \circ f(0) \end{aligned}$$

Hence $h \circ H$ is the required path-homotopy.

Consider $[f], [g] \in \pi_1(X, x_0)$. We want to show that $h_*([f]) \star h_*([g]) = h_*([f] \star [g])$.

$$h_*([f]) \star h_*([g]) = [h \circ f] \star [h \circ g]$$

Note that it suffices to show that $h(f \star g) = (h \circ f) \star (h \circ g)$. Now,

$$\begin{aligned} h(f \star g)(s) &= \begin{cases} (h \circ f)(2s), & s \in [0, \frac{1}{2}] \\ (h \circ g)(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases} \\ &= (h \circ f) \star (h \circ g) \end{aligned}$$

Hence proved. □

Theorem 2.7. If $h : (X, x_0) \rightarrow (Y, y_0)$ and $k : (Y, y_0) \rightarrow (Z, z_0)$ then

- (a) $(k \circ h)_* = k_* \circ h_*$
- (b) If $\text{id} : (X, x_0) \rightarrow (X, x_0)$ is the identity map then $(\text{id})_*$ is the identity $\pi_1(X, x_0) \rightarrow \pi_1(X, x_0)$.

Proof. (a)

$$\begin{aligned} (k \circ h)_*([f]) &= [(k \circ h) \circ (f)] \\ &= [k(h(f))] \\ &= [k_*(h_*[f])] \\ &= (k_* \circ h_*)[f] \end{aligned}$$

$$(b) (\text{id})_*[f] = [\text{id} \circ f] = [f]$$

□

Corollary 2.7.1. If $h : X \rightarrow Y$ is a homeomorphism then $\pi(X, x_0) \cong \pi(Y, h(x_0))$ via h_* .

Definition 2.8. Let $P : E \rightarrow B$ be a surjective map. A open set $U \subseteq B$ is said to be **evenly covered** by P if $P^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha}$, where V_{α} are open sets of E for all α and $P|_{V_{\alpha}}$ is a homeomorphism of V_{α} onto U .

Definition 2.9. Let $P : E \rightarrow B$ be surjective if every point $b \in B$ has a neighbourhood $U \subset B$ such that U is evenly covered by P . Then we say that P is a **covering map** and E is a **covering space** of B .

Example 2.2.

(1) $\text{id} : X \rightarrow X$.

(2) $X \times \{1, 2, \dots, n\} \xrightarrow{P} X$ where $P(X, i) = x$.

Theorem 2.10. The map $P : \mathbb{R} \rightarrow S'$ such that $x \mapsto (\cos(2\pi x), \sin(2\pi x))$ is a covering space.

Proof. $P^{-1}(U) = \bigsqcup_{n \in \mathbb{Z}} (n - \frac{1}{4}) (n + \frac{1}{4}) =: V_n$

Restricted to each V_n the map P is a homeomorphism because $\sin(2\pi x)$ is strictly monotonic on V_n for all n .

TODO: finish □

Definition 2.11. If a path-connected space X has $\pi_1(X, x_0) \cong 0$ then we say that X is **simply-connected**.

Lemma 2.12. If X is simply-connected, then any two paths in X with the same initial point and end point are path-homotopic.

Proof. $f \star \bar{g} \simeq_p e_{x_0}$ and $f \simeq_p f \star e_{x_1} \simeq_p f \star (\bar{g} \star g) \simeq_p (f \star \bar{g}) \star g \simeq_p e_{x_0} \star g \simeq_p g$. □

Definition 2.13. Let $p : E \rightarrow B$ be a surjective map, $f : X \rightarrow B$ be a map from $X \rightarrow B$. Then a map $\tilde{f} : X \rightarrow E$ is called **lifting** of f to E if $p \circ \tilde{f} = f$.

$$\begin{array}{ccc} & E & \\ & \downarrow p & \\ X & \xrightarrow{\tilde{f}} & B \end{array}$$

Lemma 2.14 (Path Lifting Lemma). Let $p : E \rightarrow B$ be a covering map with $p(e_0) = b_0$. Any path $f : [0, 1] \rightarrow B$ beginning at b_0 can be lifted uniquely to a path $\tilde{f} : [0, 1] \rightarrow E$ beginning at e_0 .

Proof. Cover B by open sets U , each of which is evenly covered by p . We can find a subdivision (partition) defined by $s_0, \dots, s_n \in [0, 1]$ such that $f([s_i, s_{i+1}])$ is contained entirely inside a evenly covered U for all i . **TODO: Think about why this is possible.** Now we will define $\tilde{f}$ inductively. Suppose that $\tilde{f}(0) = e_0$ and it has been defined for $0 \leq s \leq s_i$ for some i . We will now define $\tilde{f}$ at $[s_i, s_{i+1}]$. Suppose that $f([s_i, s_{i+1}]) \subset U$ where U is evenly covered open set of B . This implies that $p^{-1}(U) = \sqcup_{\alpha} V_{\alpha}$. Now, $\tilde{f}(s_i)$ lies in one such V_{α} say V_0 .

Define $\tilde{f}$ on $[s_i, s_{i+1}]$ as $\tilde{f}(s) := (p|_{V_0})^{-1}(f(s))$ for all $s \in [s_i, s_{i+1}]$. By pasting lemma $\tilde{f}$ is continuous on $[0, s_{i+1}]$ and we argue inductively to define $\tilde{f}$ for $[0, 1]$.

By induction, we can again suppose that for another lift $\tilde{f}' : [0, 1] \rightarrow E$ of f with $\tilde{f}'(0) = e_0$, it agrees with $\tilde{f}$ of $0 \leq s \leq s_i$ for some i

$$\tilde{f}'(s) = \tilde{f}(s), \quad \forall s \in [0, s_i]$$

Now we would like to show that

$$\tilde{f}'(s) = \tilde{f}(s), \quad \forall s \in [s_i, s_{i+1}]$$

$$\tilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s), \quad \forall s \in [s_i, s_{i+1}]$$

$\tilde{f}'(s_i)$ lies in V_0 since $\tilde{f}'(s_i) = \tilde{f}(s_i)$. $\implies \tilde{f}'([s_i, s_{i+1}])$ lies in V_0 .

$p \circ \tilde{f}' = f$ on $[s_i, s_{i+1}]$. But $\tilde{f}'([s_i, s_{i+1}]) \subset V_0$ and $p|_{V_0}$ is a homeomorphism.

Then for all $s \in [s_i, s_{i+1}]$

$$\implies \tilde{f}'(s) = (p|_{V_0})^{-1} \circ f(s) = \tilde{f}(s)$$

□

Lemma 2.15 (Path Homotopy Lifting Lemma). *Let $p : E \rightarrow B$ be a covering map. Let $p(e_0) = b_0$. Let $f : I \times I \rightarrow B$ be a map, with $f(0,0) = b_0$. Then there exists a lifting of f to E such that $\tilde{f}(0,0) = e_0$. If f is a path homotopy, so is $\tilde{f}$.*

Proof. TODO: $I \times I$ is a rectangle. Pick a partition of both sides, then the image of these small rectangles will be contained in U

1. Define $\tilde{f}$ on $0 \times I$ and $I \times 0$ by lifting both of those paths to paths that begin at e_0 .
2. Take a subdivision of $I \times I$ say $I_i \times J_j$ such that each of those small rectangles are contained in an evenly covered open set of B .
3. By induction assume that $\tilde{f}$ has been defined on $I_i \times J_j$ such that $i < i_0, j = j_0$.
4. Now, to define $\tilde{f}$ on $I_{i_0} \times I_{j_0}$ we consider the image of the union of left and bottom edge of $I_{i_0} \times I_{j_0}$ under $\tilde{f}$. This lies inside $V_0 \subset p^{-1}(U)$. □

Remark 2.16. *The lifted path homotopy is also unique.*

Lemma 2.17. *Let $P : E \rightarrow B$ be a covering map with $P(e_0) = b_0$. Let f, g be two paths in B from b_0 to b_1 . Let $\tilde{f}, \tilde{g}$ be their lifts beginning at e_0 . Then if $f \simeq_p g \implies \tilde{f}$ and $\tilde{g}$ end at the same point and they are path homotopic.*

Proof. Assume $f \simeq_p g$ with the path homotopy $F : I \times I \rightarrow B$. Then Lemma 2.15 implies that F lifts to $\tilde{F} : I \times I \rightarrow E$ with $\tilde{F}(0, t) = e_0$.

Then we can say that $\tilde{F}(1, t) = e_1$ for some $e_1 \in P^{-1}(b_1)$, $\forall t \in I$. By uniqueness of path lifting, f, g have unique lifts to E , beginning at e_0 . We also know that $\tilde{F}(s, 0)$ and $\tilde{F}(s, 1)$ are lifts of f, g both beginning at e_0 .

Then $\tilde{F}(s, 0) = \tilde{f}$ and $\tilde{F}(s, 1) = \tilde{g}$ for all $s \in I$. Hence $\tilde{F}(1, 0) = \tilde{F}(1, 1) \implies \tilde{f}(1) = \tilde{g}(1)$, and $\tilde{f} \simeq \tilde{g}$. □

Theorem 2.18. $\pi_1(S^1, (1, 0) = b_0) \cong (\mathbb{Z}, +)$

Proof.

$$\begin{array}{ccc} & & \mathbb{R} \\ & \nearrow \tilde{f} & \downarrow \\ I & \xrightarrow{f} & S^1 \\ & 7 & \end{array}$$

Consider the converging map

$$P : \mathbb{R} \rightarrow S^1, P(x) = (\cos(2\pi x), \sin(2\pi x))$$

Let $[f] \in \pi_1(S^1, b_0)$. Let $\tilde{f}$ be the lift of f to $\mathbb{R}$ beginning at 0. Now consider $\tilde{f}(1)$. It will be an integer since $\tilde{f}(1) \in P^{-1}(b_0) = P^{-1}((1, 0))$. Define

$$\phi : \pi_1(S^1, b_0) \rightarrow \mathbb{Z}, \quad \phi([f]) = f(1)$$

By previous lemma, ϕ is well-defined.

Now we show that ϕ is surjective. Let $n \in \mathbb{Z}$. Define $\tilde{f}$ to be the path in $\mathbb{R}$ which starts at 0 and ends at n . Then by definition, $(P \circ \tilde{f})$ will start at $(1, 0)$ and end at $(1, 0)$ where $\tilde{f}$ is by definition the lift of $(P \circ \tilde{f})$ beginning at 0. So $\phi([P \circ \tilde{f}]) = n$.

To show that ϕ is injective suppose that $\phi([f]) = \phi([g]) = n$ for some $[f], [g] \in \pi_1(S^1, b_0)$ we need to show that $[f] = [g] \in \pi_1(S^1, b_0)$.

Now let $\tilde{f}$ and $\tilde{g}$ be two lifts of f and g respectively beginning at 0. Moreover, since $\mathbb{R}$ is simply connected and $\tilde{f}(0) = \tilde{g}(0) = 0, \tilde{f}(1) = \tilde{g}(1) = n, \tilde{f} \simeq_p \tilde{g}$, then $P \circ \tilde{f}$ will be the path-homotopy between f and g . This implies $[f] = [g]$.

Now we want to show that ϕ is a group homomorphism. Let $[f], [g] \in \pi_1(S^1, b_0)$ such that $\phi([f]) = n, \phi([g]) = m$ for $n, m \in \mathbb{Z}$. Now we want to show that $\phi([f] \star [g]) = \phi([f \star g]) = n + m$. We need to find a lift of $f \star g$ beginning at 0 and ending at $n + m$.

Let $\tilde{f}, \tilde{g}$ be lifts of f, g respectively, $\tilde{f}(0) = \tilde{g}(0) = (0)$.

Define a path on $\mathbb{R}$ by

$$h(s) = \begin{cases} \tilde{f}(2s), & s \in [0, \frac{1}{2}] \\ n + \tilde{g}(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases}$$

Then h is a path beginning at 0 and ending at $n + m$. Now consider,

$$P(h(s)) = \begin{cases} P \circ \tilde{f}(2s), & s \in [0, \frac{1}{2}] \\ P \circ (n + \tilde{g}(2s - 1)) = P \circ \tilde{g}(2s - 1), & s \in [\frac{1}{2}, 1] \end{cases}$$

This implies that $h(s)$ is a lift of $f \star g(s) \implies \phi([f \star g]) = h(1) = n + m$ □

Corollary 2.18.1. *Fundamental Theorem of Algebra.*

Proof. Let $p(z) = z^n + a_1 z^{n-1} + \dots + a_n$, where $a_i \in \mathbb{C}$.

For the sake of contradiction assume that p has no roots in $\mathbb{C}$. Then for each $r \in \mathbb{R}^+$ define a function

$$f_r(s) = \frac{\frac{p(re^{2\pi i s})}{p(r)}}{\left| \frac{p(re^{2\pi i s})}{p(r)} \right|} \in \mathbb{C}$$

Note that $f_r(s)$ is a loop in S^1 based at $(1, 0)$ for all r . ($f_r(0) = (1, 0) = f_r(1)$ and $|f_r(s)| = 1, \forall s$). Note that as r varies f_r gives a path homotopy in S^1 between loops

that are based at $(1, 0)$. Now, note that $f_0(s)$ is the trivial loop based at $(1, 0)$. This implies,

$$[f_r(s)] = [f_0(s)] = 0 \in \pi_1(S^1, (1, 0))$$

Now fix a large value of r such that $r > |a_1| + \dots + |a_n|$ and $r > 1$. Now, we have for $|z| = r$

$$|a_1 z^{n-1} + \dots + a_n| \leq |a_1| r^{n-1} + \dots + |a_n|.$$

Thus if $r > |a_1| + \dots + |a_n|$, then

$$|z^n| = r^n > r^{n-1}(|a_1| + \dots + |a_n|) \geq |a_1 z^{n-1} + \dots + a_n|.$$

Hence,

$$|z^n| > |a_1 z^{n-1} + \dots + a_n|$$

This implies that the polynomial $p_t(z) = z^n + t(a_1 z^{n-1} + \dots + a_n)$, $\forall t \in [0, 1]$ has no roots in circle $|z| = r$. If we now replace p by p_t in the definition of $f_r(s)$ we get

$$f_r(s, t) = \frac{\frac{p_t(re^{2\pi i s})}{p_t(r)}}{\left| \frac{p_t(re^{2\pi i s})}{p_t(r)} \right|}$$

At $t = 1$,

$$f_r(s, 1) = f_r(s) \in \pi_1(S^1, (1, 0))$$

At $t = 0$,

$$f_r(s, 0) = e^{2\pi i n s}, \quad \forall s \in [0, 1]$$

So $[f_r(s, 0)] = n \in \mathbb{Z} \cong \pi_1(S^1, (1, 0))$. But $f_r(s, t)$ as we vary $t \in [0, 1]$ gives a path homotopy between $f_r(s, 0)$ and $f_r(s, 1) \implies [f_r(s, 0)] = [f_r(s)] \in \pi_1(S^1, (1, 0)) \implies n = 0$ contradicting that p is not constant. $\square$

Remark 2.19. In general, we have $P : (E, e_0) \rightarrow (B, b_0)$ be a covering map. If E is path-connected then there exists a surjection $\phi : \pi_1(B, b_0) \rightarrow P^{-1}(b_0)$. If E is simply-connected, then ϕ is a bijection.

Proof. HW1 iirc $\square$

Theorem 2.20.

$$\pi_1(\mathbb{R}^2 \setminus \{(0, 0)\}, b_0) \cong \mathbb{Z} \cong \pi_1(S^1), \quad b_0 \in S^1.$$

Proof. Let $x_0 \in S^1$. Then the inclusion map

$$J : (S^1, b_0) \rightarrow (\mathbb{R}^2 \setminus \{(0, 0)\}, b_0)$$

induces an isomorphism

$$J_* : \pi_1(S^1, b_0) \rightarrow \pi_1(\mathbb{R}^2 \setminus \{(0, 0)\}, b_0).$$

Consider

$$r : \mathbb{R}^2 \rightarrow S^1, \quad r(x) = \frac{x}{\|x\|}.$$

Observe that $r \circ J = \text{id}$, thus

$$r_* \circ J_* = \text{id}.$$

Thus J_* is injective.

Choose $[f] \in \pi_1(\mathbb{R}^2 \setminus \{(0,0)\}, b_0)$. Recall that,

$$J_* \circ r_*([f])$$

is explicitly $J \circ r \circ f$. Let

$$g(s) = \frac{f(s)}{\|f(s)\|}.$$

Next show $g \simeq_p f$. Define

$$F : I^2 \rightarrow \mathbb{R}^2 \setminus \{(0,0)\}$$

such that

$$F(s, t) = t \cdot \frac{f(s)}{\|f(s)\|} + (1-t)f(s).$$

F is the straight line homotopy and does not cross 0.

Since $f(s) \neq 0$ by definition, suppose

$$t \cdot \frac{1}{\|f(s)\|} + (1-t) = 0$$

implies $\|f(s)\|$ negative or $t \notin [0, 1]$.

Thus F is well-defined, since $f \simeq_p g$,

$$J_* \circ r_*([f]) = [f].$$

Thus surjection. □

3. RETRACTIONS

Definition 3.1. If A is a subspace of X s.t. $\exists$ a map

$$r : X \rightarrow A$$

with $r(a) = a$ for all $a \in A$, then we say A is a retract of X and r is a retraction map from X to A .

Definition 3.2. Let A be a subspace of X . A is a deformation retract of X if there exists a homotopy

$$H : X \times I \rightarrow X$$

such that

$$H(x, 0) = x \quad \forall x \in X,$$

$$H(x, 1) \in A,$$

$$H(a, t) = a.$$

Theorem 3.3. Let A be a deformation retract of X . Then $j : (A, x_0) \hookrightarrow (X, x_0)$ induces an isomorphism in $\pi_1(A, x_0)$ to $\pi_1(X, x_0)$. If A is a retract of X , then j is injective.

Proof. **TODO: hw**

□

Remark 3.4. If X deformation retracts to A , then $\pi_1(X) \cong \pi_1(A)$.

Remark 3.5. Let $\iota : A \hookrightarrow X$ then $\iota_* : \pi_1(A) \rightarrow \pi_1(X)$ is injective if X only retracts to A .
TODO: only retracts??

4. APPLICATIONS OF π_1 -COMPUTATIONS

Theorem 4.1 (Brouwer-Fixed Point Theorem). *Every map $h : D^2 \rightarrow D^2$ has a fixed point, i.e. $h(x) = x$ for at least one $x \in D^2$.*

Proof. Suppose towards contradiction $h(x) \neq x$ for all $x \in D^2$. Define a function $\gamma : D^2 \rightarrow S^1$ as follows: consider the straight line starting at $h(x)$ and going to x . Define $r(x)$ as the intersection of that line with $\partial D^2 = S^1$. Now r is a continuous function, and $r|_{S^1} = Id_{S^1}$. That is, r is a retraction of D^2 to S^1 .

Since for the inclusion $\iota : S^1 \hookrightarrow D^2$ we should have $\pi_1(S^1, (1,0)) \xrightarrow{\iota_*} \pi_1(D^2, (1,0))$ is an injection which gives a contradiction. $\square$

Theorem 4.2 (Borsuk-Ulam Theorem). *Every map $f : S^2 \rightarrow \mathbb{R}^2$ has a pair of anti-podal points (x is an antipode of $-x$, $x \in S^2$) in S^2 such that $f(x) = f(-x)$.*

Proof. Suppose that this is false. Then $f(x) \neq f(-x)$ for all $x \in S^2$. We define $g : S^2 \rightarrow S^1$ by

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$$

Define a loop η by the equator that lies in the $X - Y$ plane of S^2 .

$$\eta(s) = (\cos(2\pi s), \sin(2\pi s), 0), s \in [0, 1]$$

and let $h : I \rightarrow S^1$ defined by $h := g \circ \eta$. Note that since $g(x) = -g(-x)$ we have $h(s + \frac{1}{2}) = -h(s)$ for all $s \in [0, \frac{1}{2}]$.

$$\begin{aligned} g(\cos(2\pi(s + 1/2)), \sin(2\pi(s + 1/2)), 0) &= g(-\cos(2\pi s), -\sin(2\pi s), 0) \\ &= -g(\cos(2\pi s), \sin(2\pi s), 0) \end{aligned}$$

By path-lifting lemma, we can lift h to a path in $\mathbb{R}$, say $\tilde{h}$ starting at 0 such that

$$\tilde{h}\left(s + \frac{1}{2}\right) = \tilde{h}(s) + \frac{q(s)}{2}$$

for some odd integer values $q(s)$. But now, $q(s) = 2(\tilde{h}(s + \frac{1}{2}) - \tilde{h}(s))$ and hence it is also a continuous function. This implies $q(s)$ is a constant map (since the domain of q is connected), i.e. $\tilde{h}(s + \frac{1}{2}) = \tilde{h}(s) + \frac{q}{2}$.

$$\tilde{h}(1) = \tilde{h}\left(\frac{1}{2}\right) + \frac{q}{2} = \tilde{h}(0) + q$$

This implies $\tilde{h}(1) = q$ and hence $[h] \in \pi_1(S^1, (1,0))$ is not homotopic to the identity. But since η is nullhomotopic in S^2 , $g \circ \eta$ is homotopic to a constant map $[h] = 0 \in \pi_1(S^1, (1,0))$. A contradiction! $\square$

5. HOMOTOPY EQUIVALENCES

Theorem 5.1. $\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$ for any pair of topological spaces X and Y .

Proof. Let $P_1 : X \times Y \rightarrow X$ and $P_2 : X \times Y \rightarrow Y$ be the respective projections.

$$\phi : \pi_1(X \times Y, x_0 \times y_0) \rightarrow \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

where

$$\phi([f]) = (P_1)_*[f] \times (P_2)_*[f]$$

ϕ is surjective: Let $[g] \in \pi_1(X, x_0)$ and $[h] \in \pi_1(Y, y_0)$. Then define $[g] \times [h] : I \rightarrow X \times Y$ by $s \mapsto g(s) \times h(s)$. Then $\phi([g] \times [h]) = [g] \times [h]$.

ϕ is injective: Suppose that $[f] \in \pi_1(X \times Y, x_0 \times y_0)$ such that $\phi([f]) = (P_1)_*([f]) \times (P_2)_*([f]) = 0$. Then $P_1 \circ f \simeq_p e_{x_0}$ and $P_2 \circ f \simeq_p e_{y_0}$ by homotopies F_t and G_t .

Then, $F_t \times G_t : I \rightarrow X \times Y$ gives $f \simeq_p e_{x_0 \times y_0} \implies [f] = 0$.

$$\implies \pi_1(S^1 \times S^1) \cong \mathbb{Z} \times \mathbb{Z}. \quad \square$$

Definition 5.2. We say $f : X \rightarrow Y$ is a homotopy equivalence if there exists a $g : Y \rightarrow X$ such that $f \circ g \simeq id_Y$ and $g \circ f \simeq id_X$. Then we say that X is homotopy equivalent to Y and write it as $X \simeq_h Y$.

Remark 5.3. Homotopy equivalence is an equivalence relation on topological spaces.

Proposition 5.4. Suppose that X deformation retracts to A . Then $X \simeq_h A$.

Proof. Let $H_t : X \rightarrow A$ be the deformation retraction. Then consider $H_1(X) \subseteq A$, and $H_1|_A = id$. Let us call $r := H_1$ and let $\iota : A \hookrightarrow X$ be the inclusion map. Then $r \simeq id \implies \iota \circ r \simeq id$ and $r \circ \iota = id \implies X \simeq_h A$. $\square$

Remark 5.5. Note not all homotopy equivalence come from deformation retractions.

Remark 5.6. π_1 is not only an invariant of homeomorphism class of a topological space but also an invariant of the homotopy equivalence class of the space.

Theorem 5.7. Let $\phi : X \rightarrow Y$ be a homotopy equivalence. Then $\phi_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, \phi(x_0))$ is an isomorphism.

Lemma 5.8. If $\phi_t : X \rightarrow Y$ be a homotopy and h is the path given by $\phi_t(x_0)$ for a choice of $x_0 \in X$, then we have $(\phi_0)_* = \widehat{h}((\phi_1)_*)$ on $\pi_1(X, x_0)$. That is,

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{(\phi_1)_*} & \pi_1(Y, \phi_1(x_0)) \\ & \searrow (\phi_0)_* & \downarrow \widehat{h} \\ & & \pi_1(Y, \phi_0(x_0)) \end{array}$$

Proof. Let $[f] \in \pi_1(X, x_0)$. Let h_t be a reparameterization of the path h such that $h_t(s) := h(st)$. In particular, h_t is a path from $\phi_0(x_0)$ to $\phi_t(x_0)$. Now the composition, $h_t \star \phi_t \circ f \star \bar{h}_t$ gives a path homotopy between

$$h_0 \star \phi_0 \circ f \star \bar{h}_0 = e_{x_0} \star \phi_0 \circ f \star \bar{e}_{x_0} \simeq \phi_0 \circ f$$

and

$$h_1 \star \phi_1 \circ f \star \bar{h}_1 = h \star \phi_0 \circ f \star \bar{h} \implies (\phi_0)_* = \widehat{h}((\phi_1)_*)$$

□

Theorem 5.9. *Let $\phi : X \rightarrow Y$ be a homotopy equivalence. Then $\phi_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, \phi(x_0))$ is an isomorphism.*

Proof. Since ϕ is a homotopic equivalence, there exists a $\psi : Y \rightarrow X$ such that $\phi \circ \psi \simeq \text{id}_Y$ and $\psi \circ \phi \simeq \text{id}_X$. Consider,

$$\pi_1(X, x_0) \xrightarrow{\phi_*} \pi_1(Y, \phi(x_0)) \xrightarrow{\psi_*} \pi_1(X, \psi(\phi(x_0))) \xrightarrow{\phi_*} \pi_1(Y, \phi(\psi(\phi(x_0))))$$

Note that $\psi_* \circ \phi_* = (\psi \circ \phi)_* = \widehat{h} \circ (\text{id}_X)_*$ for some path h in X (by the previous lemma). This implies $\psi_* \circ \phi_* = \widehat{h} \implies \psi_*$ is surjective since $\widehat{h}$ is an isomorphism.

ψ_* is injective because

$$\phi_* \circ \psi_* = \widehat{h'} \circ (\text{id}_Y)_*$$

for some path h' in Y (by the previous lemma).

Hence, $\phi_* = (\psi_*)^{-1} \circ \widehat{h}$ which is an isomorphism.

□

6. $\pi_1(S_n)$

Lemma 6.1. *If a space X is the union of a collection of path-connected open sets A_α each containing the base point $x_0 \in X$. Let $A_\alpha \cap A_\beta$ be path connected for any α, β , then every loop in X based at x_0 is path homotopic to a composition of loops each contained in a single A_α for some α and based at x_0 .*

Proof. Let $[f] \in \pi_1(X, x_0)$. Suppose that $0 = S_0 < S_1 < \dots < S_m = 1$ is a subdivision of $[0, 1]$ such that $f|_{[S_{i-1}, S_i]}$ is contained in a single A_α for all i and some α depending on i . Let us denote that A_α by A_i .

Note $f(0) = x_0$. Define g_i to be the path from x_0 to $f(S_i) \in A_i \cap A_{i+1}$.

Consider,

$$(f \star \overline{g_1}) \star (g_1 \star f \star \overline{g_2}) \star (g_2 \star f \star \overline{g_3}) \star \dots \star (g_{m-1} \star f) \simeq_p f$$

while each of them are loops based at x_0 in A_i s as i varies. □

Theorem 6.2. $\pi_1(S^n, x_0) \cong \{0\}$ if $n \geq 2$.

Proof. On S^2 consider two open sets $A_1 = S^2 \setminus \{\text{north pole}\}$ and $A_2 = S^2 \setminus \{\text{south pole}\}$. Let $[f] \in \pi_1(S^2, x_0)$. Then by the lemma $f \simeq_p g_1 \star g_2$ where $[g_1] \in \pi_1(A_1, x_0)$ and $[g_2] \in \pi_1(A_2, x_0)$ for some g_1 and g_2 loops based at x_0 . Then,

$$[f] = 0 \in \pi_1(S^2, x_0) \implies \pi_1(S^2, x_0) = 0$$

□

Remark 6.3 (Stereographic Projection). $S^n \setminus \{\text{some pt}\}$ is homeomorphic to $\mathbb{R}^n$.

7. COVERING SPACES (AGAIN)

Definition 7.1. **TODO:** $\mathbb{R}P^n := S^n / \{x \sim -x\}$ (quotient space)

Theorem 7.2. Define $p : S^n \rightarrow \mathbb{R}P^n, x \mapsto [x]$. Then p is a covering map.

Proof. First show that p is an open map (i.e. U open $\implies p(U)$ open). Let $V \subset S^n$ be an open set (S^n has subspace topology from $\mathbb{R}^{n+1}$). Consider ‘antipodal map’,

$$a : S^n \rightarrow S^n, \quad x \mapsto -x$$

Then a is a homeomorphism. Thus its inverse is continuous, and $a(V)$ is open in S^n . Now since $p^{-1}(p(V)) = V \cup a(V)$, which is open in S^n , p is an open map (using definition of quotient topology).

Now we will show p is a covering map. Let $y \in \mathbb{R}P^n$. Pick $x \in p^{-1}(y)$. Choose a small neighborhood U of x in S^n such that U contains no antipodal points. Then $p|_U : U \rightarrow p(U)$ is a homeomorphism (since p is open and continuous and a bijection).

Similarly, $p|_{a(U)} : a(U) \rightarrow p(a(U))$ is also an homeomorphism. Note $p(a(U)) = p(U)$ which is an open set around $y \in \mathbb{R}P^n$. So each point in $\mathbb{R}P^n$ is evenly covered by an open set in $\mathbb{R}P^n$. Hence, p is a covering map (specifically a 2-fold covering).

Then $\pi_1(\mathbb{R}P^n, x_0) \hookrightarrow p^{-1}(x_0) \implies |\pi_1(\mathbb{R}P^n, x_0)| = 2 \implies |\pi_1(\mathbb{R}P^n, x_0)| = \mathbb{Z}/2\mathbb{Z}$. □

Corollary 7.2.1. $\mathbb{R}P^n$ is not homotopy equivalent to S^n for $n \geq 2$.

Proof. Fundamental group is a homotopy-equivalence invariant, and S^n for $n \geq 2$ is simply connected. $\square$

Remark 7.3. Homeomorphism $X \cong Y \implies X \simeq_h Y$ (homotopy equivalence).

Theorem 7.4. $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$.

Proof. Suppose for contradiction there exists a homeomorphism f from $\mathbb{R}^2 \rightarrow \mathbb{R}^n$. Then $f|_{\mathbb{R}^2 \setminus \{pt\}}: \mathbb{R}^2 \setminus \{x\} \rightarrow \mathbb{R}^n \setminus \{f(x)\}$ is also a homeomorphism. So $\mathbb{R}^2 \setminus \{x\} \simeq_h \mathbb{R}^n \setminus \{f(x)\}$ must be homotopy equivalent, But π_1 is a homotopy equivalence so we should have $\pi_1(\mathbb{R}^2 \setminus \{x\}) \cong \pi_1(\mathbb{R}^n \setminus \{f(x)\}) \implies \mathbb{Z} \cong 0$. A contradiction! $\square$

Theorem 7.5. Fundamental group of the 'figure eight space' is not abelian.

Proof. Let X be the figure eight space. Then $X = S^1 \sqcup S^1 \setminus (1,0) \cup (1,0)$ with the quotient space topology. Let x_0 be the base point. Define a covering space $E \subseteq \mathbb{R}^2$, where $E = \{x\text{-axis}\} \cup \{y\text{-axis}\} \cup \{\text{one circle tangent to each the } x\text{-axis and } y\text{-axis at each nonzero integer point}\}$ (fig 2.1 in munkres). We claim that there exists a projection map $p: E \rightarrow X$. Consider two paths $\tilde{f}, \tilde{g}: I \rightarrow E$ defined by $\tilde{f}(s) = (s, 0)$, $\tilde{g}(s) = (0, s)$. Define $f := p \circ \tilde{f}$, $g := p \circ \tilde{g}$. That is f wraps once around A , g wraps once around B .

Claim: $[f \star g] \neq [g \star f] \in \pi_1(\infty, x_0)$.

Proof: Claim $\iff f \star g \not\simeq_p g \star f$. For the sake of contradiction suppose $f \star g \simeq_p g \star f$. Now consider the lift of $f \star g$ beginning at $(0,0)$ to E given by $\tilde{f} \star B_1$, and the lift of $g \star f$ beginning at $(0,0)$ to E given by $\tilde{g} \star A_1$. If they are path-homotopic, they should have the same end points, but $(\tilde{f} \star B_1)(1) = (1,0) \neq (0,1) = (\tilde{g} \star A_1)(1)$. $\square$

Corollary 7.5.1. $\pi^2 \# \pi^2 \not\cong S^2$. $\mathbb{R}P^2$, where π^2 is the hollow torus, and $\#$ is the connected sum.

Proposition 7.6. Let $h: S^1 \rightarrow Y$ where Y is some topological space, then h is homotopic to a constant map iff h can be extended to a continuous map $D^2 \rightarrow Y$.

Proof. ($\implies$)

Let $H: S^1 \times I \rightarrow Y$ be a homotopy from h to a constant map $S^1 \rightarrow x_0$. Define $\pi: S^1 \times I \rightarrow D^2$ by $\pi(x, t) = (1-t)x$. Then $\pi|_{S^1 \times [0,1)}$ is a bijection to $D^2 \setminus \{0\}$ and it maps $S^1 \times \{1\}$ to $0 \in D^2$. Since $S \times I$ is compact and D^2 is Hausdorff, π is a closed map. Now H induces a map $g: D^2 \rightarrow Y$ such that $g \circ \pi = H$, given by

$$g(x) = \begin{cases} H(\pi^{-1}(x)) & x \neq 0 \\ x_0 & x = 0 \end{cases}$$

Then g will be the extension we want ... $\square$

TODO: 05/04 notes

8. CLASSIFICATION OF COVERING SPACE

Definition 8.1. We say that a covering space $\tilde{X}$ of X is a universal cover if $\pi_1(\tilde{X}) = 0$.

Example 8.1.

- (1) $\mathbb{R}$ is a universal cover of S^1 .
- (2) Every simply connected space is a universal cover of itself.
- (3) $\mathbb{C} \rightarrow \mathbb{C}$ where $z \mapsto z^2$.
- (4) S^n is a universal cover of $\mathbb{R}P^n$ for $n \geq 2$

Remark 8.2. An object with property x is universal if any other object with property x factors through the first object.

Theorem 8.3. Suppose that $\tilde{X}$ is a universal cover of X , then if X is path-connected and locally-connected, any other path-connected cover X_1 of X , $\tilde{X}$ covers X_1 . Moreover the following diagram commutes.

$$\begin{array}{ccccc} \tilde{X} & \xrightarrow{\quad p_1 \quad} & X_1 & \xrightarrow{\quad p_2 \quad} & X \\ & \searrow \quad \quad \quad \nearrow & & & \\ & & p & & \end{array}$$

Proof. The proof follows from the following two lemmas. □

Lemma 8.4. Suppose that $X, X_1, \tilde{X}$ are all path-connected such that $X_1 \xrightarrow{p_1} X$, $\tilde{X} \xrightarrow{p} X$ are covers with $\pi_1(\tilde{X}) = 0$ then $\exists p_2 : \tilde{X} \rightarrow X_1$ such that $p_1 \circ p_2 = p$.

Proof. Follows from the general lifting proposition. □

Lemma 8.5. Let $X, X_1, \tilde{X}, p, p_1, p_2$ be defined as before, then p_2 is a covering map.

Proof. Let $x_1 \in X_1$ be a point, U be a neighborhood of $p_1(x_1) \in X$ that is evenly covered by both p_1 and p . Now let $p_1^{-1}(U) = \sqcup_{\alpha} U_{\alpha}$ and $p^{-1}(U) = \sqcup_{\alpha'} V_{\alpha'}$. Moreover assume that V is in the slice of $p_1^{-1}(U)$ where x_1 lies. Claim: V is evenly covered by p_2 . Proof: Consider the U_{α} 's in $\tilde{X}$ that are mapped to V . Lets say that collection is $\{U_{\beta}\}$. Then $p_2^{-1}(V) = \sqcup_{\beta} U_{\beta}$ and $p_2|_{U_{\beta}}$ is a homeomorphism since it is a composition of $p|_{U_{\beta}}$ and $(p_1|_V)^{-1}$ which are both homeomorphisms □

Theorem 8.6. Let $\tilde{X}$ be a universal cover of X , then $\tilde{X}$ is unique upto homeomorphism.

Proof. Hint: Lifts are unique if we fix a base point of the lift.

Let $\tilde{X}$ and $\bar{X}$ be two universal covers of X . Then,

$$\begin{array}{ccc}
 & (\bar{X}, \bar{x}_0) & \\
 f \nearrow & \downarrow p_2 & \\
 (\tilde{X}, \tilde{x}_0) & \xrightarrow{p_1} & (X, x_0)
 \end{array}$$

$$\begin{array}{ccc}
 & (\tilde{X}, \tilde{x}_0) & \\
 g \nearrow & \downarrow p_1 & \\
 (\bar{X}, \bar{x}_0) & \xrightarrow{p_2} & (X, x_0)
 \end{array}$$

$$p_1 = p_2 \circ f, \quad p_2 = p_1 \circ g$$

That is,

$$p_1 = p_1 \circ g \circ f$$

Moreover, $g \circ f(\tilde{x}_0) = \tilde{x}_0$. Hence, $g \circ f$ is a lift of p_1 to $\tilde{X}$ by p_1 . Identity is also such a lift, then $g \circ f = \text{id}$ (by uniqueness of lift homotopy). $\square$

TODO: Read this proof ?? For any sub space of $\pi_1(X) \leftrightarrow$ cover X_1 of X with $p_*(\pi_1(X_1)) \cong$ that subspace.

Remark 8.7. Any collection of homomorphisms $\phi_\alpha : G_\alpha \rightarrow H$ extends uniquely to $\phi : \star_\alpha G_\alpha \rightarrow H$ such that $\phi(g_1 \cdots g_n) \mapsto \phi_{\alpha_1}(g_1)\phi_{\alpha_2}(g_2) \cdots \phi_{\alpha_n}(g_n)$.

where G_α is a free group, H is a group, and $\star_\alpha G_\alpha$ consists of all words of arbitrary length where each letter belongs to a G_{α_i} and consecutive letters are from different G_{α_i} .

Now suppose that X is decomposed as a union of path-connected spaces $\{A_\alpha\}$ each of which contains the base point $x_0 \in X$. Then inclusion $A_\alpha \hookrightarrow X \implies j_\alpha : \pi_1(A_\alpha) \rightarrow \pi_1(X)$ induce $\Phi : \star_\alpha \pi_1(A_\alpha) \rightarrow \pi_1(X)$.

Let $(\iota_{\alpha\beta})_* : \pi_1(A_\alpha \cap A_\beta) \rightarrow \pi_1(A_\alpha)$ be induced from inclusion. Hence,

$$(j_\alpha)_* \circ (\iota_{\alpha\beta})_* = (j_\beta)_* \circ (\iota_{\beta\alpha})_* : \pi_1(A_\alpha \cap A_\beta) \rightarrow \pi_1(X)$$

If $w \in \pi_1(A_\alpha \cap A_\beta)$ then $(\iota_{\alpha\beta})_*(w) ((\iota_{\beta\alpha})_*(w))^{-1}$ is in the kernel of Φ .

Let N denote the normal subgroup of $\star_\alpha \pi_1(A_\alpha)$ generated by such elements. Then $\ker(\Phi) = N$ (under assumptions).

Theorem 8.8 (Seifert-Van Kampen Theorem). If X is a union of path-connected open sets A_α such that $x_0 \in X$, and $x_0 \in A_\alpha$ for all α , and $A_\alpha \cap A_\beta$ is path-connected, and $A_\alpha \cap A_\beta \cap A_\gamma$ is also path-connected for all α, β, γ . Then Φ induces an isomorphism

$$\pi_1(X) \cong \star_\alpha \pi_1(A_\alpha) / N$$

Proof. We have shown before that Φ is surjective. Now we will show that $\ker(\Phi) = N$. Define a factorization of an element $[f] \in \pi_1(X)$ as the product $[f_1] \dots [f_k]$ such that each $[f_i] \in \pi_1(A_\alpha)$ for some A_α and $f \sim_p f_1 \star f_2 \star \dots \star f_k$. A factorization of f is thus a word in $\star_\alpha \pi_1(A_\alpha)$. We will say that any two such factorizations are equivalent if they are related by the following two moves or their inverses:

- (1) Combine adjacent terms $[f_i][f_{i+1}]$ into a single term if $[f_i][f_{i+1}]$ are in the same group.
- (2) If $f_i \in A_\alpha \cap A_\beta$ we may regard $[f_i] \in \pi_1(A_\alpha)$ as $[f_i] \in \pi_1(A_\beta)$.

The first move does not change the factorization. The second move does not the element in $\star_\alpha \pi_1(A_\alpha) / N$.

Let us assume that $[f_1] \dots [f_k]$ and $[f'_1] \dots [f'_l]$ are two factorizations of $[f] \in \pi_1(X)$. We will show that they are equivalent. We have,

$$f_1 \star \dots \star f_k \cong f'_1 \star \dots \star f'_l$$

That is, there exists a path-homotopy $F : I \times I \rightarrow X$. Now choose a partition of $I \times I$ by

$$0 = s_0 < \dots < s_m = 1 \text{ and } 0 = t_0 < \dots < t_n = 1$$

such that each rectangle $R_{ij} = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ is mapped to a single A_α by F . We label this A_α by A_{ij} .

Now we will modify the partition in two ways:

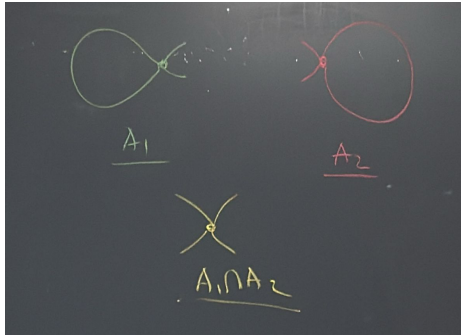
- (1) Add subdivisions to the top ($[t_{n-1}, t_n]$) and bottom row ($[t_0, t_1]$) such that for each end point in the partition there exists a corresponding s_i .
- (2) Have at least 3 rows.
- (3) Every point in $I \times I$ must lie in at most 3 (smaller) rectangles
- (4) Rename the smaller rectangles (left to right, bottom to top) using R_i .

Define γ_r as the path from the left to the right edge of $I \times I$, and it separates rectangles $R_1, \dots, R_r$ from the rest of the rectangles.

Note that F_{γ_r} is a loop based at x_0 . And all F_{γ_r} are path homotopic. Then $F|_{\gamma_r}$ has a factorization. If $F(v) \neq x_0$, then we have a path from $x_0 \rightarrow F(v) = F|_{\gamma_1}$ $\square$

Example 8.2.

- (1) $\pi_1(\infty, x_0)$ (∞ denotes the figure 8 space).



Then, $\pi_1(\infty, x_0) \cong (\pi(A_1) \times \pi_1(A_2))/N = (\mathbb{Z} \times \mathbb{Z})/N$. Since $N = A_1 \cap A_2$ deformation retracts to a point this is $\mathbb{Z} \times \mathbb{Z}$.

(2) $\pi_1(S^1 \wedge S^2) \cong \mathbb{Z}$

$X_1 \wedge X_2 := (X_1 \sqcup X_2)/(x_1 \sim x_2) \text{ for some } x_1 \in X_1, x_2 \in X_2.$

(3) $\pi_1(\mathbb{R}^3 \setminus S^1)$ **TODO:**

9. CELL COMPLEXES

Definition 9.1. We start with a discrete space whose points are regarded as 0-cells. Now, we inductively define the *n-skeleton* X^n from X^{n-1} by attaching n -cells e_α^n via maps $\phi_\alpha : S^{n-1} \rightarrow X^{n-1}$ where e_α^n s are D_α^n . Hence, $X^n = X^{n-1} \sqcup D_\alpha^n / (x \sim \phi_\alpha(x) \text{ if } \alpha, \forall x \in S^{n-1})$. We say a topological space X has a CW/CX structure if $X = \bigcup_n X^n$.

Definition 9.2. Each cell e_α^n of a CW CX has a map $\Phi_\alpha : D_\alpha^n \rightarrow X$ which extends $\phi_\alpha : \partial D_\alpha^n \rightarrow X$. It is defined by

$$D_\alpha^n \hookrightarrow X^{n-1} \sqcup D_\alpha^n \xrightarrow{\phi_\alpha : \partial D_\alpha^n \rightarrow X^{n-1}} X^{n-1} \sqcup D_\alpha^n = X^n \hookrightarrow X$$

This map is called the **characteristic map** associated to ϕ_α .

Definition 9.3. A **subcomplex** A of a CW CX X is a subspace $A \subset X$ such that A is a union of some number of cells in X with a attaching map for each of those cells has a image contained in A . We say that (X, A) is a CW pair.

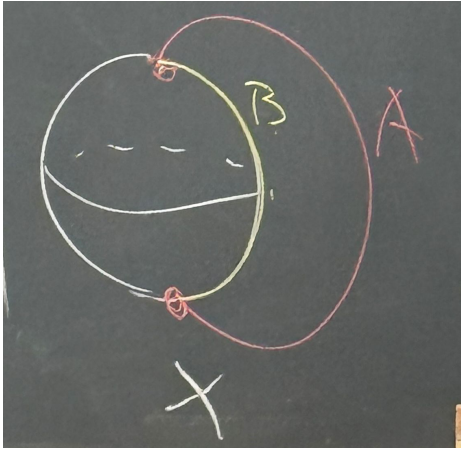
Definition 9.4 (Quotient CX). If (X, A) is a CW pair, then X/A inherits a canonical cell structure from X . The cells of X/A are the cells $X - A$, plus one 0-cell given by the image of A in X/A . For a cell e_α^n of $X - A$ with $\phi_\alpha : S^{n-1} \rightarrow X^{n-1}$, the corresponding attaching map in X/A is

$$S^{n-1} \rightarrow X^{n-1} \rightarrow X^{n-1}/A^{n-1}$$

Theorem 9.5. If (X, A) is a CW pair and A is contractible then $q : X \rightarrow X/A$ is a homotopy equivalence.

Proof. Hatcher Chapter 0, Homotopy Extension Lemma □

Example 9.1. (1) Consider a space X built from S^2 by attaching an arc to two points in S^2 .



Both (X, A) and (X, B) are CW complexes.

(2) $\pi_1(S^3, S^1) = ?$ **TODO:**

10. HOMOLOGY

Remark 10.1. $\pi_n(S^n) = \mathbb{Z}$, and $\pi_m(S^n) = 0$ if $m > n$. We don't know how to compute $\pi_m(S^n)$ for $m < n$.

Homology distinguishes higher dimension space and it is easy to compute. We define simplicial and singular homology separately, but eventually they have the same properties.

10.1. Simplicial Homology.

Definition 10.2. A n -**simplex** is the smallest convex set in $\mathbb{R}^m$ spanned by $(n + 1)$ points $v_0, \dots, v_n$ such that all of them do not lie in any hyperplane of dimension less than n . That is, $v_1 - v_0, v_2 - v_0, \dots, v_n - v_0$ are linearly independent. We say $(v_0, \dots, v_n)$ comes with the order $v_0, \dots, v_n$.

The standard n -simplex is

$$\left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid \sum_i t_i = 1, t_i \geq 0 \right\}$$

There is a canonical homeomorphism from Δ^n to any n -simplex.

$$(t_0, \dots, t_n) \mapsto \sum t_i v_i$$

Definition 10.3. Δ -complex structure of a space X is a collection of maps $\sigma_\alpha : \Delta^n \rightarrow X$.

- (1) $\sigma_\alpha|_{\Delta_n^\circ}$ is injective and each point of X is in the image of exactly one such $\sigma_\alpha|_{\Delta_n^\circ}$.
- (2) $\sigma_\alpha|_{\text{face of } \Delta_n}$ is one of the maps $\sigma_\beta : \Delta^{n-1} \rightarrow X$, where a face of Δ^n is a $(n - 1)$ -simplex that we get by deleting a vertex of Δ^n .
- (3) A set $U \subset X$ is open iff $\sigma_\alpha^{-1}(U)$ is open in Δ^n , for all α .

Definition 10.4. Let X be a Δ -complex and let $\Delta_n(X)$ denote the free abelian group with basis n -simplices of X . The elements of $\Delta_n(X)$ are of the form $\sum_i n_i \sigma_i$ where this is a finite sum, $n_i \in \mathbb{Z}$ and $\sigma_i : \Delta^n \rightarrow X$.

We denote the deletion of v_i from the simplex as $[v_0, \dots, \widehat{v}_i, \dots, v_n] \cong [v_0, \dots, v_{n-1}]$. We will always orient the 1-simplex $[v_i, v_j]$ by an arrow going from v_i to v_j for $i < j$.

Proposition 10.5. The composition

$$\Delta_n(X) \xrightarrow{\partial_n} \Delta_{n-1}(X) \xrightarrow{\partial_{n-1}} \Delta_{n-2}(X)$$

is zero ($\partial_{n-1} \circ \partial_n = 0$).

Proof.

$$\begin{aligned}\partial_n(\sigma) &= \sum_i (-1)^i \sigma|_{[v_0, \dots, \widehat{v}_i, \dots, v_n]} \\ \partial_{n-1} \circ \partial_n(\sigma) &= \sum_{j < i} (-1)^i (-1)^j \sigma|_{[v_0, \dots, \widehat{v}_j, \dots, \widehat{v}_i, \dots, v_n]} + \sum_{i < j} (-1)^i (-1)^{j-1} \sigma|_{[v_0, \dots, \widehat{v}_i, \dots, \widehat{v}_j, \dots, v_n]}\end{aligned}$$

□

Definition 10.6. In general, whenever we have (C_n, ∂_n)

$$\rightarrow C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}}$$

with $\partial_n \circ \partial_{n+1} = 0$, then $\text{img}(\partial_{n+1}) \subset \ker(\partial_n)$. Hence we define the n -th homology group,

$$H_n := \ker(\partial_n) / \text{im}(\partial_{n+1})$$

Hence the n -th simplicial homology of X is $H_n^\Delta(X) = \ker(\partial_n) / \text{im}(\partial_{n+1})$.

Example 10.1. Let $X = S^1$. There is one 0-simplex v and one 1-simplex e . Then,

$$\Delta_0(S^1) = \mathbb{Z}\langle v \rangle, \Delta_1(S^1) = \mathbb{Z}\langle e \rangle, \Delta_2(S^1) = 0$$

And we have,

$$0 \xrightarrow{\partial_2} \Delta_1(S^1) \xrightarrow{\partial_1} \Delta_0(S^1) \xrightarrow{\partial_0} 0$$

with

$$H_0^\Delta(S^1) = \frac{\ker(\partial_0)}{\text{im}(\partial_1)} = \frac{\mathbb{Z}\langle v \rangle}{\sigma_1|_{[v_1]} - \sigma_1|_{[v_0]}} = \frac{\mathbb{Z}\langle v \rangle}{0} \cong \mathbb{Z}\langle v \rangle \cong \mathbb{Z},$$

$$H_1^\Delta(S^1) = \frac{\ker(\partial_1)}{\text{im}(\partial_2)} = \frac{\mathbb{Z}\langle v \rangle}{0} \cong \mathbb{Z}\langle v \rangle \cong \mathbb{Z},$$

and

$$H_2^\Delta(S^1) = \frac{\ker(\partial_2)}{\text{im}(\partial_3)} = 0$$

We conclude that $H_n^\Delta(S^1) \cong \mathbb{Z}$ if $n = 0, 1$ and $H_n^\Delta(S^1) = 0$ otherwise.

Example 10.2.

$$\begin{array}{ccc} v_0 & \xrightarrow{a} & v_0 \\ b \uparrow & \nearrow^L c & \uparrow b \\ v_0 & \xrightarrow{a} & v_0 \end{array}$$

We have,

- $\Delta_0(T^2) = \mathbb{Z}\langle v_0 \rangle$
- $\Delta_1(T^2) = \mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle \oplus \mathbb{Z}\langle c \rangle$
- $\Delta_2(T^2) = \mathbb{Z}\langle U \rangle \oplus \mathbb{Z}\langle L \rangle$

- $\Delta_3(T^2) = 0$

$$0 \xrightarrow{\partial_3} \Delta_2(T^2) \xrightarrow{\partial_2} \Delta_1(T^2) \xrightarrow{\partial_1} \Delta_0(T^2) \xrightarrow{\partial_0} 0$$

The homologies are,

- $H_0^\Delta(T^2) = \frac{\ker(\partial_0)}{\text{im}(\partial_1)} = \frac{\mathbb{Z}\langle v_0 \rangle}{0} \cong \mathbb{Z}\langle v_0 \rangle \cong \mathbb{Z}$
- $H_1^\Delta(T^2) = \frac{\ker(\partial_1)}{\text{im}(\partial_2)} = \frac{\mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle \oplus \mathbb{Z}\langle c \rangle}{\mathbb{Z}\langle a+b-c \rangle} \cong \frac{\langle a, b, c \rangle}{\langle a+b-c \rangle} \cong \langle a, b \rangle \cong \mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle$
- $H_2^\Delta(T_2) = \frac{\ker(\partial_2)}{\text{im}(\partial_3)} = \frac{\mathbb{Z}\langle U-L \rangle}{0} \cong \mathbb{Z}\langle U-L \rangle \cong \mathbb{Z}$
- $H_n^\Delta = 0$ if $n \geq 3$.

Example 10.3. Compute $H_n^\Delta(\mathbb{R}P^2)$.

$$\begin{array}{ccc} v_0 & \xleftarrow{b} & v_1 \\ a \uparrow & U & \nearrow c \\ v_1 & \xrightarrow{b} & v_0 \\ & L & \downarrow a \end{array}$$

TODO: Try; Hatcher has this worked out

Definition 10.7. A **singular n -simplex** in a space X is a map $\sigma : \Delta^n \rightarrow X$, where there are no conditions of $\text{im}(\sigma)$ restricted to a face.

Definition 10.8. A **chain complex** is defined as the (finite) formal sum of singular n -simplices.

$$C_n(X) := \left\{ \sum_i n_i \sigma_i \mid \sigma_i : \Delta^n \rightarrow X \right\}$$

with the differential $\partial_n : C_n(X) \rightarrow C_{n-1}(X)$ where,

$$\partial_n(\sigma) := \sum_i (-1)^i \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, v_n]}$$

and $\partial_n \circ \partial_{n+1} = 0$. The **n -th singular homology** is

$$H_n(X) := \frac{\ker(\partial_n)}{\text{im}(\partial_{n+1})}$$

Remark 10.9. $H_n(X)$ is an invariant under the homeomorphism class of X .

Proposition 10.10. Let $X = \sqcup X_\alpha$ then $H_n(X) \cong \oplus_\alpha H_n(X_\alpha)$, where each X_α is a distinct path connected component of X .

Proof. We have $C_n(X) \cong \oplus_\alpha C_n(X_\alpha)$ and $\partial_n = \oplus_\alpha \partial_n^\alpha$ which implies

$$H_n(X) \cong \oplus_\alpha H_n(X_\alpha)$$

□

Proposition 10.11. If X is non-empty and path-connected then $H_0(X) \cong \mathbb{Z}$.

Proof. By definition,

$$H_0(X) \cong \frac{C_0(X)}{\text{im}(\partial_1)}$$

We define a homomorphism

$$\varepsilon : C_0(X) \rightarrow \mathbb{Z}, \quad \sum_i n_i \sigma_i \mapsto \sum_i n_i$$

This map is surjective since for any $n \in \mathbb{Z}$, $\varepsilon(n\sigma_0) = n$ where $\sigma_0 : \Delta^0 \rightarrow X$.

Claim: $\ker(\varepsilon) = \text{im}(\partial_1)$.

Proof. Note that $\text{im}(\partial_1) \subseteq \ker(\varepsilon)$ since for a singular 1-simplex $\sigma : \Delta^1 \rightarrow X$ we have $\varepsilon(\partial_1(\sigma)) = 0$.

Now for $\ker(\varepsilon) \subseteq \text{im}(\partial_1)$, let $\varepsilon(\sum_i n_i \sigma_i) = 0 \implies \sum_i n_i = 0$.

Choose paths $\tau : I \rightarrow X$ for $i > 0$, from a base point $x_0 \in X$ to $\sigma_i(v_0)$ and define $\sigma_0(v_0) = x_0$. We can view $\tau_i : [v_0, v_i] \rightarrow X$ as a singular 1-simplex. Thus, $\delta(\tau_i) = \sigma_i(v_0) - x_0$. Hence,

$$\partial \left(\sum_i n_i \tau_i \right) = \sum_i n_i \sigma_i(v_0) - \sum_i n_i \cdot x_0 = \sum_i n_i \sigma_i(v_0) \implies \sum_i n_i \sigma_i \in \text{im}(\partial_1)$$

Thus $\ker(\varepsilon) = \text{im}(\partial_1)$. □

That is,

$$H_0(X) \cong \frac{C_0(X)}{\text{im}(\partial_1)} = \frac{C_0(X)}{\ker(\varepsilon)} \cong \mathbb{Z}$$

□

Proposition 10.12. *If X is a point, then $H_n(X) \cong 0$ for $n > 0$ and $H_0(X) \cong \mathbb{Z}$.*

Proof. **TODO:** □

Definition 10.13. *We will change the definition of homology so that a point has trivial $H_0(-)$. The **reduced homology groups** $\widetilde{H}_n(X)$ is defined to be the homology of the following chain complex.*

$$\rightarrow C_n(X) \xrightarrow{\partial_n} C_{n-1}(X) \xrightarrow{\partial_{n-1}} \dots \xrightarrow{\partial_1} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \xrightarrow{0} 0$$

So, $\widetilde{H}_n(X) \cong H_n(X)$ if $n \geq 1$. Moreover we have $\text{im}(\partial_1) \subseteq \ker(\varepsilon)$

TODO: *i missed this part*

$$H_0(X) \cong \widetilde{H}_0(X) \oplus \mathbb{Z} \implies \widetilde{H}_0(pt) = 0$$